

# Kaushik R Bangera

Software Engineer | Backend Developer

Over 2 years of experience working with Node.js, MySQL and AWS

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## Experience

### Juego Studios – Mangalore

Software Engineer – 2+ Years, Aug 2021 – Jan 2024

- Created a **highly resilient and scalable Ingestion System** on AWS, utilizing **Kinesis**.  
Reduced estimated costs by ~66%
- Structured **Glue Catalog tables**, reducing query time **from 7 seconds to 1-2 seconds**
- Designed and executed a **data transformation pipeline** to cleanse and convert raw data through **Glue ETL jobs**
- Implemented various AWS services using Terraform, adhering to **Infrastructure as Code** principles
- Integrated **Agones** into a project as a Proof of Concept (PoC), harnessing its capabilities as an **open-source, multiplayer dedicated game server** scaling tool that operates on **Kubernetes**
- Developed **M2M (Machine-to-Machine) Authentication** architectures
- Employed **SQL transactions** and optimized database performance, enhancing data retrieval with **Redis**
- Achieved **1000 CCU and 1000 requests per second** by modifying buffer sizes in the database (**Aurora MySQL**)
- Deployed and managed **game servers** using **Kubernetes**
- Established **JWT token-based authentication** using **Access and Refresh tokens**
- Securely integrated **Metamask** wallet and **Keplr** wallet
- **Collaborated with cross-functional teams**, including frontend, gameplay, CMS, and game design teams
- Provided **mentorship to team members** and delegated tasks effectively.

## Internal Innovation

- Developed a **Lambda RAM picker** to execute Lambda functions with various RAM configurations, determining the **most cost-effective RAM allocation**
- Reused existing EC2 as **NAT instance** to replace **NAT Gateway**, reducing cost
- Conducted extensive research and development on **Amazon Personalize**
- Created a room handler featuring a **bucket system** to refine server-side game loops
- Established a **caching strategy** for master tables
- Implemented **database synchronization** to maintain consistent table structure

## Skills

- Node.js
- JavaScript
- MySQL
- Python
- WebSocket
- Ingestion System
- ETL
- AWS
  - Lambda
  - EC2
  - S3
  - Aurora MySQL
  - API Gateway
  - Kinesis Data Stream
  - Kinesis Firehose
  - Step Function
  - Glue
  - Athena
  - SQS
  - Cognito
  - Amazon Personalize
  - Cloud Watch
- Docker
- MiniKube
- Kubernetes
- Agones
- Terraform
- Git
- PySpark
- MongoDB
- Metamask
- Keplr
- NGINX

# Projects

## **Pixelynx** (Node.js, Python, Terraform, Lambda, Kinesis, Glue Job, S3, Step function and other AWS services)

- Designed a **highly available ingestion system** on AWS. Optimized and **reduced estimated costs** from ~\$1200 to ~\$400.
- Developed API structures and **Infrastructure as Code (IaC)** using **Terraform** for AWS services.
- Implemented **M2M Auth architecture** for authentication and created a **blockchain scraper** using **Moralis**.
- Designed a **data transformation pipeline** using **Glue ETL** and **PySpark** to convert to **Parquet format**, reducing storage and enhancing query speed.
- Responsible for architecture design, task distribution, and providing progress updates to clients.

## **Super Rummy** (Node.js, Lambda, MySQL, Redis, EC2, API Gateway )

- **Estimated efforts** for implementation, collaborated with frontend, gameplay, CMS teams, and game designer.
- Added **bots** when there are no players to match.
- Optimized and implemented a **bucket system** for idle/timed-out rooms.

## **Stemuli** (Node.js, MySQL, Azure Kubernetes Service, Azure CDN, Azure Virtual Machine )

- Designed API and database structures, utilizing **SQL transactions** for matchmaking.
- Deployed **game servers with Kubernetes**, implementing scaling based requests for room.
- Led architecture design, **mentored the team**, and managed **task distribution**.

## **Retro Hunter** ( Node.js, Lambda, MySQL, Redis, Metamask wallet )

- Ensured secure authentication using the **Metamask** wallet with nonce and signature verification.
- Optimized the codebase for enhanced performance and implemented **data caching strategies**.
- **Collaborated** with the frontend team to resolve issues and optimize code.
- Responsibilities included **bug fixing, feature implementation, and optimization**.

## **Strange Clan – an RPG game:** (Node.js, Lambda, MySQL, Ganache, Keplr wallet)

- Designed sprint plan, **API, and database structure** design
- Integrated **Keplr** wallet for authentication and **Ganache** for local testing
- Implemented a secure way of authenticating using a nonce and signature
- Designed an **efficient quest system** and created a marketplace for trading NFTs
- Responsibilities included bug fixing, feature implementation, and optimization

## **Poker, Spintopia** (Node.js, Lambda, Aurora MySQL, Redis, API Gateway )

- **Refactored the codebase**, rewrote game logic to match **Texas Hold'em** rules
- Designed **API and database structure**, optimized data retrieval using Redis.
- Estimated efforts, created a **sprint plan**, and implemented **JWT token-based authentication**.
- Created **role-based permissions**.
- Integrated **Cognito Authentication** and **Secrets Manager** service.

# Education

## **Srinivas Institute of Technology**

B.E. in Computer Science Engineering - ( Aug 2017 - Jul 2021 )